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COMMUNICATION

Lewis acid-catalyzed unexpected selective C–C bond cleavage: an efficient and mild construction of cyclopentenes†

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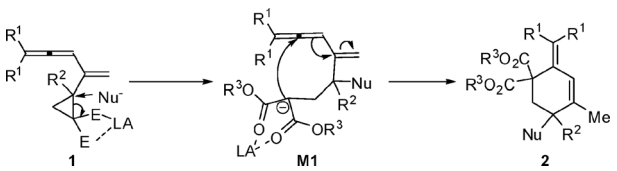
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An intriguing Lewis acid-catalyzed intramolecular ring-opening/cyclization reaction of cyclopropylvinylallenes is developed. This process proceeds at ambient temperature to afford the structurally diverse cyclopentenes exclusively, which demonstrates the unique reactivity of 1,3,4-alkatrien-2-yl cyclopropanes.

Selective cleavage of C–C bonds and subsequent incorporation into construction of complex molecules represents a fundamental challenge in organic chemistry. Carbon–carbon bonds in cyclopropanes are generally much more reactive because of their unique strained three-membered carbocyclic structure,[1] therefore, they play a crucial role as unique units in modern synthetic chemistry and have attracted the interest of organic chemists for many years.[1b–e,2] Lewis acid (LA) promoted [3 + *n*] annulations of donor–acceptor cyclopropanes have received considerable attention in recent years: a variety of hetero- or carbocyclic products have been constructed.[3–8] The Lewis acid-catalyzed intramolecular [3 + 2] cycloadditions of cyclopropanyl 1,1-dicarboxylate are less prevalent than an intermolecular variant despite the potential in the synthesis of natural products. Snider *et al.* reported a preliminary result on an intramolecular [3 + 2] cycloaddition of a cyclopropane 1,1-diester with a C=C bond.[9a] Recently, Kerr *et al.* reported an intramolecular [3 + 2] cycloaddition of a cyclopropane 1,1-diester with a C=N bond by which aza-[3.3.0] and aza-[4.3.0] skeletons were constructed;[9b,c] these reactions have been successfully applied to the total synthesis of FR901483 and allosecurinine.[9e,f] Wang *et al.* reported their results on the construction of the bridged oxa- and aza-[*n*.2.1] (*n* = 2, 3, 4) skeletons through intramolecular [3 + 2] cycloaddition of cyclopropane 1,1-diester.[9d]

On the other hand, allenes are a class of compounds of current interest with unique reactivity. During the last few decades, allenes have been recognized as a class of very useful compounds for organic synthesis.[10,11] To the best of our knowledge, the chemistry of a combination of allenyl and cyclopropane functionalities has not been reported even though it may be of great interest in organic synthesis. Recently, our group has disclosed an efficient method for the synthesis of allenes with a cyclopropylvinyl substituent.[12] This reaction proceeds smoothly under neutral conditions, with the exclusive formation of 1,3,4-alkatrien-2-yl cyclopropane derivatives.



Scheme 1 Assumption of the reaction.

Due to the unique structure of **1**, we envisioned that the cyclopropanyl carboxylate moiety in **1** should undergo ring-opening to form intermediate **M1** with the assistance of Lewis acids and nucleophiles. Due to the presence of a vicinal vinylallene substituent, the intermediate **M1** should further undergo cyclization to form the cyclohexene product **2** (Scheme 1). As continuous effort on the allene chemistry, we report our recent results on the unexpected $Zn(OTf)_2$-catalyzed ring-opening/cyclization reaction of cyclopropyl vinylallenes.

To test our hypothesis, we started the initial investigation on the reaction of 1,1-bis(methoxycarbonyl)-2-hexyl-2-(5′-ethyl-1′,3′,4′-heptatrien-2′-yl)cyclopropane **1a** with 1.2 equiv. of TMSCl in the presence of 10 mol% $FeCl_3$ in CH_2Cl_2 at 30 °C. The reaction reached completion within 1 h and delivered a product in a good yield. To our surprise, cyclopentene product **3a** was obtained instead of **2**-type cyclohexene based on the ^1H and ^{13}C spectra of the product. Then attention was paid to the optimization of the reaction conditions for this transformation. It was found that the reaction could not proceed to afford the product **3a** in most of the tested solvents, such as CH_3CN, DCE, THF, DMF, DMSO, CH_3NO_2, dioxane, 1,1,1-trichloroethane or 1,1,2-trichloroethane (Table S1, entries 2–10, ESI†). Dichloromethane was found to be the best solvent (Table S1, entry 1, ESI†).

Next, various Lewis acids were examined in this reaction (Table 1). Cyclopropane **1a** was found to be successfully converted to **3a** in moderate to good yields in the presence of Lewis acids,[13] such as $FeCl_3$, $Sc(OTf)_3$, $Zn(OTf)_2$, AgOTf, $Cu(OTf)_2$, or $ZnCl_2$ (Table 1, entries 1–6). Of these $Zn(OTf)_2$

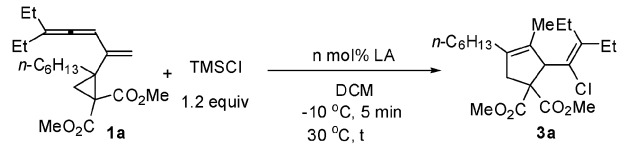
[a] *Shanghai Key Laboratory of Green Chemistry and Chemical Process, Department of Chemistry, East China Normal University, 3663 North Zhongshan Road, Shanghai 200062, P. R. China. E-mail: masm@sioc.ac.cn; Fax: +86-21-6260-9305*
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† Electronic supplementary information (ESI) available: Table S1, experimental details and analytical data of all the products, NMR spectra of **3a–3n**. CCDC 890419. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/c2cc36672j

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Table 1 Effects of Lewis acids[a]



Entry	LA (*n* mol%)	*t* (h)	Yield of **3a**[b] (%)
1	$FeCl_3$ (10)	1	75
2	$Sc(OTf)_3$ (10)	1.5	74
3	$Zn(OTf)_2$ (10)	1.5	81
4	AgOTf (10)	2.3	61
5	$Cu(OTf)_2$ (10)	2	73
6	$ZnCl_2$ (10)	24	71[d] (10)[d,e]
7	$TiCl_4$ (10)	30	0
8	$BF_3.Et_2O$ (10)	30	33[d] (23)[d,e]
9	$Zn(OTf)_2$ (5)	2	94
10	—	27	0[d]
11[c]	$Zn(OTf)_2$ (5)	27	0[d] (100)[d,e]

[a] Unless otherwise specified, the reaction was carried out using **1a** (0.2 mmol), TMSCl (31 μL, d = 0.846 g mL^{-1}, 0.24 mmol), and *n* mol% of Lewis acid in 5 mL of CH_2Cl_2 under a nitrogen atmosphere. [b] Isolated yield. [c] The reaction was carried out under a nitrogen atmosphere in the absence of TMSCl. [d] NMR yield determined by ^1H NMR analysis using 1,3,5-trimethylbenzene as the internal standard. [e] The number in parentheses is the recovery of **1a**.

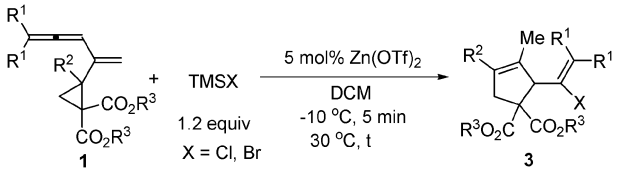
gave the best result, affording **3a** in 81% yield. In contrast, $TiCl_4$ or $BF_3{\cdot}Et_2O$[13] could not lead to the formation of **3a** (Table 1, entries 7 and 8). Interestingly, reducing the catalyst loading of $Zn(OTf)_2$ from 10 mol% to 5 mol% improved the yield of **3a** from 84% to 94% within 2 h (Table 1, entry 9). No product was obtained either in the absence of a Lewis acid (Table 1, entry 10) or TMSCl (Table 1, entry 11). Thus, we defined the reaction of 1 equiv. **1** and 1.2 equiv. TMSCl catalyzed by 5 mol% $Zn(OTf)_2$ in CH_2Cl_2 at 30 °C as the standard conditions.

With the optimized protocol in hand, we turned to demonstrate the generality of this reaction. The results summarized in Table 2 show that this reaction indeed provides a straightforward entry to a variety of cyclopentene derivatives in moderate to good yields. Different substitutions at the terminal position of allene, such as methyl, ethyl, cyclohexyl and cyclopentyl could be accommodated in the reaction, affording the corresponding cyclopentenes **3a–3n** in good yields (Table 2). Dimethyl, diethyl, and dibenzyl malonates are all proper substrates for this reaction, affording the corresponding products in moderate to good yields. Alkyl, benzyl, and allyl substituted 1,3,4-alkatrien-2-ylcyclopropanes proceeded smoothly to afford the corresponding products **3a–3n** in moderate to good yields. When 1.7 g of **1a** (4.7 mmol) was used, **3a** was obtained in 83% yield.

Product **3l** was treated with sodium hydroxide (2 M aqueous solution) in ethanol at 60 °C for 1 hour, followed by quenching with hydrochloride to afford a solid product **4** in 94% isolated yield (Scheme 2). Thus, the structure of **3** was then further confirmed unambiguously by the X-ray diffraction study of **4** (Scheme 2).[14]

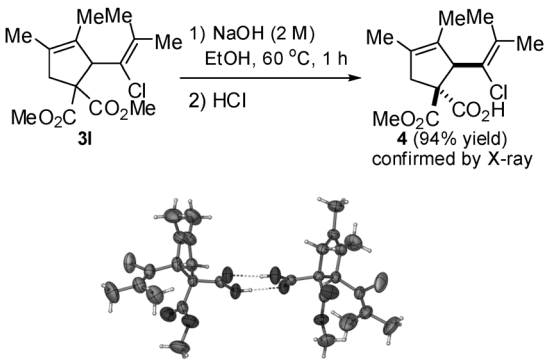
A rationale is proposed as shown in Scheme 3. Initially, cyclopropane is activated by the chelation of Lewis acids to the carbonyl groups of the ester groups to polarize the cyclopropanyl ring. Then ring-opening occurred to afford 1,3-zwitterionic intermediate **B**. The substituted group at the

Table 2 $Zn(OTf)_2$-catalyzed ring opening/isomerization/cyclization of vinylallenyl cyclopropanes[a]



Entry	R^1	R^2	R^3	*t* (h)	Yield of **3**[b] (%)
1	Et	*n*-C_6H_{13}	Me (**1a**)	2	94 (83)[c] (**3a**)
2	Et	*n*-C_4H_9	Me (**1b**)	5	87 (**3b**)
3	Et	Bn	Me (**1c**)	3	81 (**3c**)
4[d]	Et	Et	Me (**1d**)	3.5	89 (**3d**)
5	Et	$C_{10}H_{21}$	Me (**1e**)	6	96 (**3e**)
6	Et	Allyl	Me (**1f**)	4	44 (**3f**)
7[d]	Et	CH_3	Me (**1g**)	2	74 (**3g**)
8[d]	Et	*n*-C_6H_{13}	Et (**1h**)	4	90 (**3h**)
9	Et	*n*-C_6H_{13}	Bn (**1i**)	4.5	76 (**3i**)
10	–$(CH_2)_5$–	*n*-C_6H_{13}	Me (**1j**)	5.5	74 (**3j**)
11	–$(CH_2)_4$–	*n*-C_6H_{13}	Me (**1k**)	5	50 (**3k**)
12	Me	Me	Me (**1l**)	3	79 (**3l**)
13	–$(CH_2)_5$–	Bn	Me (**1m**)	12	62 (**3m**)
14[e]	Et	*n*-C_4H_9	Me (**1b**)	7	62 (**3n**)

[a] Unless otherwise specified, the reaction was carried out using **1** (0.2 mmol), TMSCl (0.24 mmol) and 5 mol% $Zn(OTf)_2$ in 5 mL of CH_2Cl_2 under a nitrogen atmosphere. [b] Isolated yield. [c] The reaction was also carried out using **1a** (4.7 mmol, 1.7014 g), TMSCl (5.64 mol) and 5 mol% $Zn(OTf)_2$ in 35 mL of CH_2Cl_2. [d] The reaction was carried out using 6 mol% $Zn(OTf)_2$. [e] The reaction was carried out using TMSBr (31.5 μL, d = 1.16 g mL^{-1}, 0.24 mmol). In **3n**, X = Br.

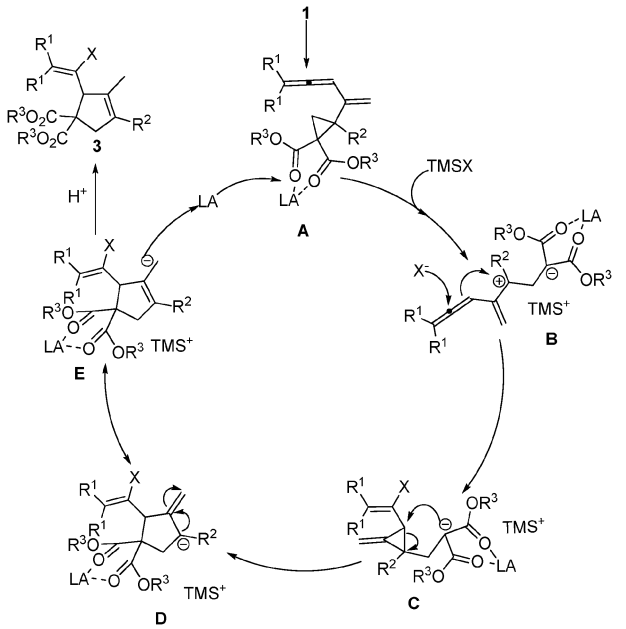


Scheme 2 Hydrolysis of **3l** and ORTEP representation of **4**.

2-position of 1,3,4-alkatrien-2-yl cyclopropane was essential to the ring-opening process of cyclopropylvinylallenes since the reaction was completely shut down when R^2 was a proton, probably due to the instability of the carbocation after ring-opening reaction. Then nucleophile X^- attacks the central carbon atom of the allene moiety, which is followed by intramolecular cyclopropanation to form intermediate **C** with a cyclopropanylmethylene unit. The carbon anion center in intermediate **C** attacks the three-membered ring to form cyclopentane intermediate **D**, which is in resonance with intermediate **E**. Finally, intermediate **E** is quenched by a proton from the environment to afford the product **3** while regenerating the Lewis acid catalyst.

In summary, we have developed a unique ring-opening/cyclization of cyclopropylvinylallenes with TMSX in the presence of a Lewis acid, which provides a straightforward

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Scheme 3 Plausible mechanism of the reaction.

way to functionalized cyclopentenes. The presence of vinylallenyl group is essential to the success of this transformation. Further studies to develop the asymmetric version and further synthetic utility of the method are underway.

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14 Crystal data for compound **4**: $C_{14}H_{19}ClO_4$, M_W = 286.74, monoclinic, space group $P2_1/c$, final R indices [$I > 2\sigma(I)$], R_1 = 0.0693, wR_2 = 0.1889, R indices (all data) R_1 = 0.0964, wR_2 = 0.2146, a = 10.4900(3) Å, b = 14.2863(4) Å, c = 20.8403(6) Å, α = 90°, β = 100°, γ = 90°, V = 3077.5(15) Å^3, T = 296(2) K, Z = 8, reflections collected/unique 35 326/5410 (R_{int} = 0.0459), number of observations [$I > 2\sigma(I)$] 3814, parameters: 343. CCDC 890419†.

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